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Experimental measurement of hydrosystem-induced delayed mortality in juvenile Snake River spring Chinook salmon (*Oncorhynchus tshawytscha*) using a large-scale acoustic array

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Abstract: Out-migrating Snake River salmon smolts must pass eight major hydro dams before reaching the Pacific Ocean. Direct mortality at the dams is generally low; however, the cumulative stress caused by dam passage is hypothesized to result in delayed mortality, which occurs beyond the impounded section of the river. We tested the delayed mortality hypothesis by comparing in-river and early ocean survival of hatchery-origin spring Chinook salmon (*Oncorhynchus tshawytscha*) from the Snake River to a mid-Columbia River population that passes through only four dams and has higher smolt to adult return rates. Smolts >140 mm fork length were implanted with acoustic transmitters and tracked with the Pacific Ocean Shelf Tracking (POST) array to as far as Alaska. There was no detectable difference in survivorship to the first ocean detection line, 274 km beyond the final dam (S_{Snake} = 29% ± 4%, S_{Yakima} = 28% ± 5%), indicating that the survival disparity observed in adult return rates may develop later in the marine life history phase. Our study is the first to estimate survival in the coastal ocean and demonstrates the utility of a large-scale array in testing previously intractable hypotheses.

Résumé : Les saumoneaux qui émigrent de la rivière Snake doivent traverser huit barrages hydroélectriques importants avant de rejoindre le Pacifique. La mortalité directe aux barrages est généralement basse; cependant, nous avançons l'hypothèse selon laquelle le stress cumulatif dû au passage des barrages cause une mortalité différée qui se produit au-delà la section de la rivière pourvue de barrages. Nous avons testé l'hypothèse de la mortalité différée en comparant la survie dans la rivière et au début de la période océanique chez des saumons chinook (*Oncorhynchus tshawytscha*) printaniers de pisciculture avec celle d'une population de la région médiane du Columbia qui ne traverse que quatre barrages et qui possède un taux de retour de saumoneaux à adultes plus élevé. Nous avons implanté des émetteurs acoustiques à des saumoneaux de longueur à la fourche >140 mm et avons suivi les poissons par l'intermédiaire du réseau POST (surveillance de la plate-forme du Pacifique) parfois jusqu'en Alaska. Il n'y a pas de différence décelable de survie au niveau de la première ligne océanique de détection à 274 km au-delà du dernier barrage (S_{Snake} = 29% ± 4%, S_{Yakima} = 28% ± 5%), ce qui indique que la différence de survie observée dans les taux de retour des adultes peut se développer plus tard durant la phase marine du cycle biologique. Notre étude est la première à estimer la survie dans l'océan côtier et elle démontre l'utilité d'un réseau à grande échelle pour tester des hypothèses jusqu'à maintenant irréductibles.

[Traduit par la Rédaction]

Introduction

Out-migrating Snake River Chinook salmon (*Oncorhynchus tshawytscha*) smolts must pass four major hydro dams in the Snake River and four in the lower Columbia River before reaching the Pacific Ocean. Although direct mortality at the dams has decreased in recent years (Muir et al. 2001), subsequent survival from out-migration until adult return is substantially lower for wild and hatchery Snake River spring

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Chinook salmon relative to some mainstem Columbia River populations (Berggren et al. 2008; Yakama Nation 2008). It has thus been hypothesized that delayed mortality may occur after passage out of the hydropower system (i.e., the impounded section of the rivers) because of the cumulative stress caused by dam passage (e.g., Budy et al. 2002; Schaller and Petrosky 2007). Therefore, to quantify delayed or indirect effects of dam passage, survival must be measured downstream of the final dam and preferably beyond fresh water.

The Pacific Ocean Shelf Tracking (POST) array (Welch et al. 2002) is a potentially continental-scale marine tracking array composed of individual acoustic receivers anchored to the ocean floor in a precise geometry. Each receiver records the date and time of uniquely coded acoustic tags passing nearby. In this study, cross-shelf ocean detection lines were located off southeast Alaska, British Columbia (around Vancouver Island), and Washington (40 km north of the Columbia River mouth), as well as in the Columbia and Snake rivers (Fig. 1). The potential tracking distance from the Snake River detection site to southeast Alaska was ~2300 km.

In 2006 we used the POST array to test the hypothesis that greater mortality is expressed in a population of hatchery-origin Snake River spring Chinook smolts after migrating through the impounded section of the river (eight dams) than that of a downstream population from the Yakima River (a tributary of the mid-Columbia River) that migrates through only four lower Columbia River dams. Smolt to adult return rates (SAR) of the Snake River population (Berggren et al. 2008) averaged only one-fifth that of the Yakima population (Yakama Nation 2008) for fish released between 1999 and 2005, and preliminary adult returns from the 2006 release year were consistent with prior years (SAR_{Yakima} = 2.7%; B. Bosch, Yakima–Klickitat Fisheries Project, 771 Pence Road, Yakima, WA 98908, USA, personal communication 2008; SAR_{Snake} = 0.38%, Berggren et al. 2008), providing a comparative test of whether this differential mortality develops shortly after they passed out of the hydrosystem. Our results provide the first estimates of survival and distribution in the ocean and substantially extend the period of life history during which is it possible to test whether delayed mortality occurs in juvenile salmon (Fig. 1).

Materials and methods

Populations used in the study

The Snake River population of spring Chinook used in this study was reared at the Dworshak National Fish Hatchery (NFH), on the Clearwater River (a tributary of the Snake River). Dworshak NFH spring Chinook from the Clearwater River are not a component of the Snake River evolutionary significant unit (ESU); however, their survival is routinely compared with other populations in the Snake River basin as well as with downstream Columbia River populations (see Berggren et al. 2008). Of the Snake River populations, Dworshak generally has the lowest SARs (Berggren et al. 2008), and therefore the difference in survival estimates should be particularly pronounced when compared with a population from outside of the Snake River basin. For logistical ease, Dworshak fish were transferred, tagged, and released at Kooskia NFH (60 km upstream of Dworshak NFH); distance to the Columbia River mouth is 870 km.

The Yakima population was reared at the Cle Elum Supplementation and Research Facility (CESRF) on the Yakima River. Migrating hatchery spring Chinook salmon released from CESRF acclimation sites were collected from the lower Yakima River at the Chandler Juvenile Monitoring Facility (CJMF; 194–249 km from the acclimation sites) and held for tagging. We collected fish at CJMF because mortality in the Yakima River has been as high as 80% in recent years (Yakama Nation 2008). Fish released from CJMF migrated 76 km down the Yakima River to enter the Columbia River just upstream from the Snake River – Columbia River confluence; distance to the Columbia River mouth is 615 km. Thus, both the Snake River and Yakima River populations shared a common migration path that passed through the four lower Columbia River dams before reaching the ocean listening line at Willapa Bay, a total distance of 910 and 655 km, respectively.

Tagging

We surgically implanted 794 smolts (two release groups of approximately 200 at each hatchery) with individually identifiable Vemco V9-6L acoustic transmitters (9 mm × 20 mm, 3.1 g in air, 2 g in water; Shad Bay, Nova Scotia, Canada). The tags were programmed to transmit a unique code at a mean interval of 60 s, and the rated battery life of the tag (4 months) was intended to exceed the likely migration time to the POST subarray in Alaska. A passive integrated transponder (PIT) tag was placed in the body cavity (through the incision) to ensure that tagged smolts were diverted back into the river at the dam bypass facilities. We set a minimum body size requirement of 140 mm fork length (FL), which meant that only the upper portion of the smolt size distribution could be tagged. The mean percent body weight of the transmitters was 8.3% (range = 2.6%–11.5%). The surgical procedure used to implant acoustic transmitters was approved by the Vancouver Island University Animal Care Committee.

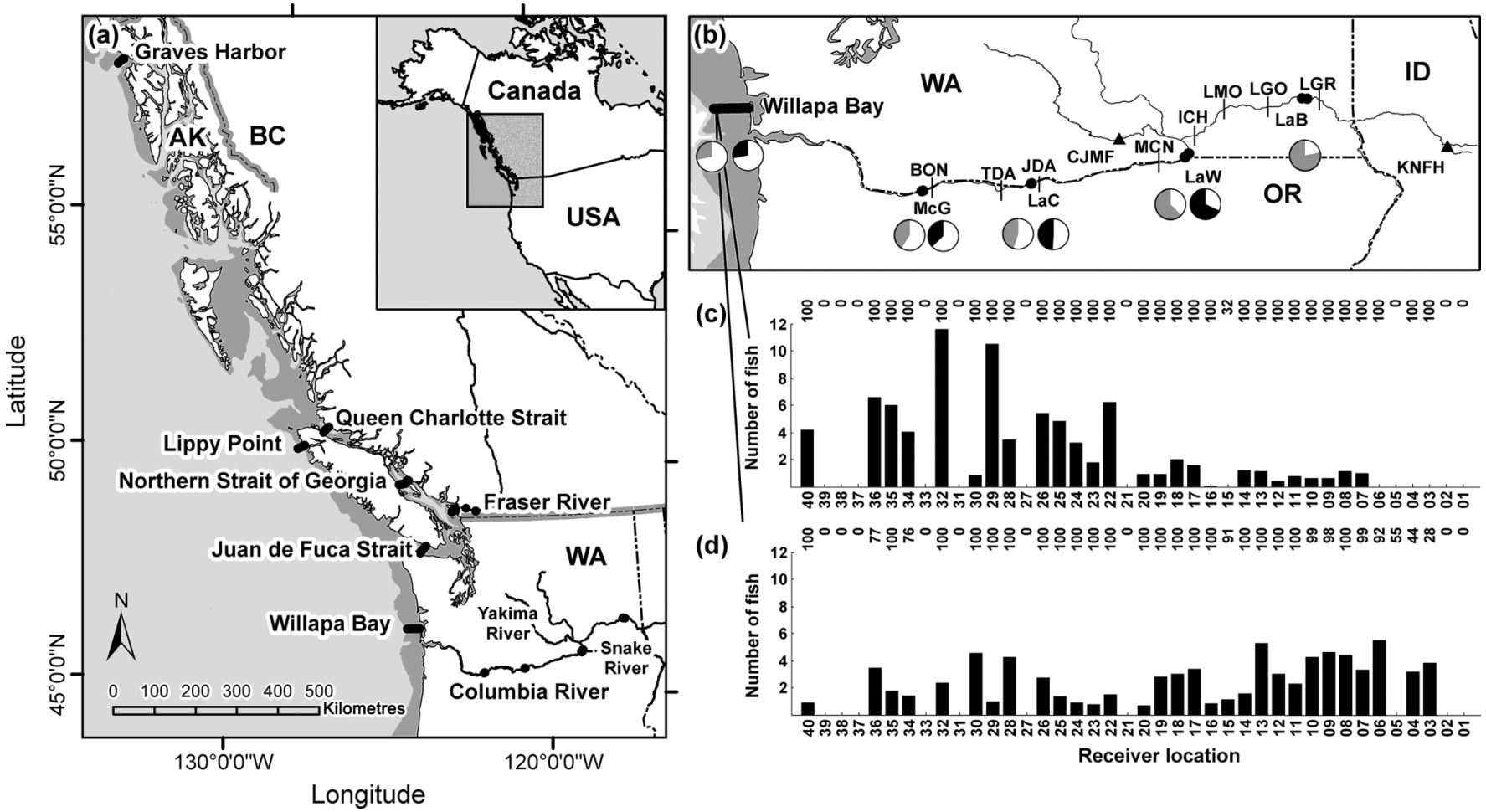
Array location

Component subarrays were deployed within the Columbia River at McGowan's Channel at river kilometre (rkm) 224 (10 km below Bonneville Dam), in Lake Celilo at rkm 340 (7 km below John Day Dam), and in Lake Wallula at rkm 502 (21 km below the confluence of the upper Columbia and Snake rivers; Fig. 1). In the Snake River, receivers were installed in Lake Bryan at Snake River rkm 159 (14 km below Lower Granite Dam; total distance from the ocean was 681 km). At each of these locations, receivers were deployed as paired lines to assist in evaluating detection probability and to provide a survival measurement below three of the major hydroelectric dams.

Oceanic components of the POST array at Willapa Bay (southern Washington), Lippy Point (northwest Vancouver Island, British Columbia), and Graves Harbor (southeast Alaska) were deployed as single lines and extended from nearshore to the edge of the continental shelf (200 m depth), a distance of up to 30 km (Fig. 1).

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Fig. 1. Panel (*a*) shows the 2006 Pacific Ocean Shelf Tracking (POST) acoustic array (black dots and thick black lines). The continental shelf (depths < 200 m) is shaded. Panel (*b*) shows the location of the release sites (CJMF, Chandler Juvenile Monitoring Facility; KNFH, Kooskia National Fish Hatchery), detection sites (McG, McGowan's Channel; LaC, Lake Celilo; LaW, Lake Wallula; LaB, Lake Bryan), and dams (BON, Bonneville; TDA, The Dalles; JDA, John Day; MCN, McNary; ICH, Ice Harbor; LMO, Lower Monumental; LGO, Little Goose; LGR, Lower Granite) within the Yakima, lower Snake, and Columbia rivers. Pie charts represent the proportion of Chinook salmon (*Oncorhynchus tshawytscha*) surviving to each detection site (Dworshak, grey; Yakima, black). Panels (*c*) and (*d*) show the cross-shelf distribution of Snake River and Yakima River smolts on the Willapa Bay line, respectively. Numbers above the bars represent the percentage of time the receiver was operational during the migration.



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Table 1. Mean survival for Snake River (Dworshak) and Yakima River (Cle Elum Supplementation and Research Facility (CESRF)) spring Chinook salmon (*Oncorhynchus tshawytscha*) implanted with V9-6L acoustic transmitters and passive integrated transponder (PIT) tags in 2006.

Detection site	Segment distance (km)	Dams passed per segment	Est. *n*	Survival probability	Cumulative survival	Detection probability
Snake River						
LaB	189	1	308	0.78±0.02	0.78±0.02	0.97±0.02
LaW	179	3	245	0.78±0.02	0.61±0.03	0.92±0.03
LaC	162	2	176	0.79±0.02	0.48±0.04	0.56±0.06
McG	116	2	162	0.82±0.06	0.40±0.05	0.72±0.07
WiB	263	0	109	0.74±0.08	0.29±0.04	0.69±0.08
Yakima River						
LaB	NA	NA	NA	NA	NA	NA
LaW	113	0	258	0.70±0.03	0.70±0.03	1.00±0
LaC	162	2	186	0.72±0.03	0.51±0.03	0.77±0.05
McG	116	2	139	0.70±0.02	0.36±0.04	0.86±0.05
WiB	263	0	106	0.78±0.10	0.28±0.05	0.69±0.08

Note: Segment distance to the first detection site is measured from release location; all others are measured from previous detection site (LaB, Lake Bryan; LaW, Lake Wallula; LaC, Lake Celilo; McG, McGowan's Channel; WiB, Willapa Bay). Survival estimates are ±standard error (SE); NA, not applicable. Snake River smolts were released on 1 May (N = 198; mean fork length (FL) = 146.9 mm, range = 140–208 mm) and 8 May (N = 198; mean FL =145.6 mm, range = 140–192 mm); Yakima River smolts released on 30 May (N = 199; mean FL = 154.5 mm, range = 140–173 mm) and 6 June (N = 199; mean FL = 154.5 mm, range = 140–168 mm). The two release dates for Yakima River smolts were chosen in an attempt to have their arrival at Bonneville Dam coincide with the Snake River release groups. Yakima River smolts entered the Columbia River upstream of confluence with Snake River and therefore were not detected on the Lake Bryan line in the Snake River.

Survival analysis

Estimates of smolt survival (Φ) and detection probability (p) at each detection site were calculated using the Cormack–Jolly–Seber (CJS) model for live recaptured animals implemented in Program MARK (White and Burnham 1999). This model jointly estimates survival and detection within a likelihood framework. Because sample size beyond Willapa Bay was low, it was not possible to reliably estimate survival to British Columbia or Alaska. Therefore, we mainly focus on estimates of survival and detection to the ocean line at Willapa Bay.

For in-river lines, we recognized basic CJS model assumptions (equal survival probability, equal recapture probability, no tag loss, and instantaneous sampling); however, for oceanic lines that are not bounded on the offshore end (continental slope), we required two additional assumptions: (*i*) as fish migrate they cross over the acoustic detection lines that span the width of the continental shelf; and (*ii*) the majority of the fish departing the Columbia River swim north. These assumptions are supported by evidence from numerous ocean sampling programs (e.g., Fisher and Pearcy 1995; Brodeur et al. 2004; Bi et al. 2007).

To estimate overdispersion in the data, we used a median $\hat{c}$ procedure (White and Burnham 1999) to test the goodness of fit (GOF) of our global model ($\Phi_{\text{population}\times\text{line}}\, p_{\text{population}\times\text{line}}$) and then corrected for this overdispersion ($\hat{c}$ = 2.1) across all candidate models. We hypothesized that distance and number of dams within a migration segment (Table 1) may affect survival, so we modeled additive effects of each external covariate ($\Phi_{\text{population+distance}}$; $\Phi_{\text{population+dam number}}$), as well as the additive effects of both covariates ($\Phi_{\text{population+distance+dam number}}$), in addition to the fully varying CJS global model. Because of the low number of fish detected at Vancouver Island and Alaska, we included in our models two additional treatment groups (each N = 100) of spring Chinook smolts from Dworshak NFH, tagged with the same acoustic tag, that were released below Bonneville Dam during the spring out-migration to better quantify the detection probability of the Willapa Bay detection line. The survival estimate of this group was modeled in the same way across models ($\Phi_{\text{group}\times\text{line}}$) and did not influence survival estimates for run-of-river smolts. Recapture parameters were modeled to vary with population and line, except at Willapa Bay where all populations were pooled.

We used an information theoretic approach (i.e., Akaike's information criterion (AIC)) to select the most parsimonious model. In general, the model with the lowest AIC value (which accounts for the number of model parameters, n_{pars}) has more support in the data, and if the ΔAIC of the other candidate models is greater than 2, then these models (i.e., hypotheses) have little or no support. We report survival estimates from the model that had the most support.

Cross-shelf distribution

For each population we used a χ^2 GOF test to determine if the cross-shelf ocean distribution adjacent to Willapa Bay was significantly different from the null hypothesis of a uniform distribution. To perform the tests, we counted the number of unique ID codes detected at each of the 40 receivers for each population. If a fish was detected at more than one receiver, a proportion was allocated to the receiver (e.g., if an ID code was detected on two receivers, each receiver was assigned a value of 0.5). If a receiver was displaced during the migration (e.g., by fishing activity), then that receiver was omitted from the data set (30 active receivers were used in the final analysis). To meet the requirements of the χ^2 test, we pooled data into 15 groups.

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Results and discussion

Survival

The most parsimonious model based on an information theoretic approach was the model that constrained survival solely as a function of distance travelled ($\Phi_{population+distance}$, AIC = 1677.0, n_{pars} = 14). Although this model had considerably more support than the global model ($\Phi_{population \times line}$, AIC = 1687.7, ΔAIC = 10.7, n_{pars} = 21), the added effect of distance on survival was negative for Dworshak smolts ($\Phi \cdot km^{-1}$ = 0.9993) as expected, but positive for the Yakima population ($\Phi \cdot km^{-1}$ = 1.0007). This was a surprising result for the Yakima population but was likely caused by low survival in the lower Yakima River (the shortest migration segment, which also lacked dams), combined with higher survival in the lower Columbia River and ocean (the longest segment). The model that constrained survival as a function of dam number also had support in the data ($\Phi_{population+dam\ number}$, AIC = 1677.2, ΔAIC = 0.2, n_{pars} = 14); however, the rate of survival per dam was slightly >1 for both populations, indicating that there was no negative survival effect due to the number of dams passed in a migration segment. The lack of effect of the number of dams passed on direct in-river survival may be a reflection of the improvements of fish passage in the hydrosystem (Muir et al. 2001), so that any remaining negative effect of the dams may be too small to be measurable once the effect of distance is accounted for. There was little support for a model that constrained survival as a function of number of dams within a migration segment and distance travelled ($\Phi_{population+distance+dam\ number}$, AIC = 1680.5, ΔAIC = 3.5, n_{pars} = 16).

Survival of Snake River smolts ranged between 74% and 82% for individual river segments and between 70% and 78% for Yakima smolts (Table 1). Survivorship of Snake and Yakima smolts to McGowan's Channel, located just below Bonneville Dam (the final dam), was 40% and 36%, respectively. There was no detectable difference in survivorship from release to Willapa Bay between populations (S_{Snake} = 29%, S_{Yakima} = 28%; Table 1), and the mean time for smolts to reach this line after passing Bonneville Dam was ~5 days for both. Therefore, delayed mortality was not evident after several days in the coastal ocean at a point located 274 km beyond the hydrosystem. Although we tagged only larger individuals, and migration time was delayed by several weeks relative to untagged runs, these survivorship estimates were very similar to independent PIT tag estimates of survival of Dworshak (see Welch et al. 2008) and Yakima (D. Lind, Yakima–Klickitat Fisheries Project, 771 Pence Road, Yakima, WA 98908, USA, personal communication 2007) spring Chinook salmon in 2006.

Tagged Chinook salmon were also detected on acoustic lines on the west coast of Vancouver Island and Alaska. The large distance between ocean detection sites (~500 and 1000 km, respectively), the relatively low numbers detected on these lines, and the fact that the detected smolts arrived at both locations within 2 days of line deployment prevented us from reliably estimating survival to these locations. However, the detection of four Snake and two Yakima smolts on the Vancouver Island line, some 1500 km distant from the release site (30 days travel time from Willapa Bay for both populations), and the detection of two Snake River smolts on the Alaska line (and none from the Yakima population), some 2500 km from the release site, is inconsistent with the delayed mortality hypothesis.

Coastal ocean distribution

The cross-shelf distribution of the Snake River population was significantly different from the null hypothesis of a uniform distribution (χ^2 GOF test; $p < 0.01$), while the Yakima population was not ($p = 0.57$; Figs. 1*a*, 1*b*), with the Snake population more concentrated on the offshore portion of the line.

Study limitations

Several factors should be considered when interpreting our data. (*i*) We were limited to tagging smolts >140 mm FL, (*ii*) tagged fish were released 2–4 weeks after the main hatchery releases, (*iii*) tagged smolts were detected on the outermost receiver at Willapa Bay, suggesting that some smolts may migrate beyond the offshore extent of the subarray, and (*iv*) arrival to the first ocean line was not simultaneous for the two populations despite our attempts to do so. Future work should address these limitations by using smaller tags, extending the marine components of the array farther offshore (to depths ≥ 200 m), and by modifying release strategies to better synchronize release dates and arrival times in the ocean.

Conclusions

Lower river and early marine survival data for Columbia River basin Chinook salmon is essential for assessing the hypothesized delayed effects of dam passage on smolt survival. In this study, we found no evidence that delayed mortality was expressed by Snake River smolts (>140 mm FL) by the time they had reached the first ocean subarray, some 274 km from Bonneville dam, when compared with a size-matched group from the Yakima River population. This result is based on the assumption that smolts turn north after ocean entry and is supported by the high proportion of smolts detected on the Willapa Bay line subsequent to passing the detection line below Bonneville Dam. The mean time of arrival from below Bonneville Dam to Willapa Bay (5 days) puts an initial lower limit on the period when differential mortality may be expressed in these populations. Our results imply that if delayed mortality causes the disparity in adult return rates of hatchery-origin Snake River spring Chinook salmon, it is at a time and place more distant from the Columbia River; however, it is plausible that delayed mortality may operate on smaller fish (<140 mm FL) that were not tested here. Our result that the number of dams passed within a migration segment did not affect survival also supports the findings that direct in-river survival of salmon smolts has increased in recent decades (Muir et al. 2001), possibly to the level of an undammed river (Welch et al. 2008). We note that our study was not designed to measure very small effects on mortality, as it was focused on making the first large-scale measurements of migratory survival for salmon originating from the Columbia River. It is not correct to conclude that the dams currently play no role in affecting survival, simply that any contributory effect appears to be small.

The differing marine distributions of the two populations at Willapa Bay provides a potential mechanism in support of the alternative hypothesis that populations from the same river basin can have different responses to ocean conditions (e.g., Levin 2003). It remains unclear whether the entire size range of these two populations, as well as wild smolts, have similar survival and behavior as the smolts reported here. The technical development of the POST array allowed us to make a direct scientific test of a key hypothesis concerning delayed mortality of a population of Snake River spring Chinook salmon in the ocean. As such, it marks an important scientific milestone. Advances in the array design, however, will allow us to study marine survival over the complete size range of naturally occurring spring Chinook smolts, as well as other endangered populations of salmon originating from the Columbia River basin.

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